

Enhancing Environmental Protection Through HCI: Insights from a Coastal Litter Cleanup Initiative in Norway

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Abstract

Norway's long coastline, industrial activities and harsh weather conditions expose it to the global environmental challenge of Marine Plastic Pollution (MPP). Ambitious grassroots coastal cleanup initiatives have been taking place across the country since 2013. The project that is the object of this case study aimed to create a unified visualization tool to enhance data-driven decision-making at local and regional levels. This involved categorizing litter to guide policy making and strengthen environmental protection efforts. However, the research team encountered significant challenge: effectively classifying litter and encouraging the decision-makers to use the tools for decision-making. A visualization tool may not be enough to foster data-driven decision-making but it may bring about the necessary collaboration among various stakeholders by providing a tangible basis and an arena to discuss this poorly-legislated problem. While the field of human-computer interaction (HCI) offers many effective practices for using technology to support environmental protection, we present a case study of an unsuccessful attempt and reflect on the lessons learnt. We focus on designing interactions for often-neglected domains, such as MPP, and reflect on adapting and developing new practices to extend HCI's research, ensuring it can accommodate the unique demands of a diverse stakeholder group and the constraints of real-world projects.

CCS Concepts

• **Human-centered computing** → **Interaction design**.

Keywords

Coastal environment, visualization, design

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1 Introduction

Norway is characterized by a long coastline, rocky shores, and often extreme weather. This creates conditions for the accumulation of marine litter, or Marine Plastic Pollution (MPP)[7]. Grassroots movements have organised throughout the country to clean up the coasts, supported by the government and private retailers' funds, as detailed in [21]. As cleanup operations are carried out across the country and the litter is categorised by volunteers and professionals, the data is recorded in a national database administered by the Environment Agency. The projects that are the object of this case study had one overarching goal, namely to bridge the volunteer community with the decision-makers at local and regional levels, by translating aggregated data into useful and intuitive visualisations. The hope was that the tools produced would help adjust local and regional environmental protection policies in an effective manner, employing targeted measures to address the sources of litter, prevent or at least mitigate leakages into the marine and coastal environments[22].

Many challenges emerged along the way, and in this paper, we focus on three main problems and the lessons learnt from the design and evaluation processes. The first challenge was to develop a usable mobile application for the volunteers to capture photos and record the litter as they proceed with clean-up operations. The second was to classify the litter found in a meaningful way for all stakeholders. And the third was to use the outputs to encourage decision-makers to introduce preventive strategies and mitigation measures for MPP.

Applying HCI insights to environmental protection or sustainable development is not a completely novel idea. Sustainable development is defined as development that satisfies the needs of the current without compromising the needs of future generations [34], and therefore has environmental protection at its heart. Sustainability science, on the other hand, is defined as problem-driven research that ties knowledge to effective actions and society benefits [33]. Research on sustainability within HCI is rich and diverse, focusing on technology and design that can contribute to environmental and societal well-being. The topics are also broad, such as sustainable interaction, green IT and energy efficiency[10], cultural and behavioral change[23], sustainability in HCI research[20]. For example, Blevis[5] discusses how design decisions in HCI can support sustainability by focusing on the lifecycle of products, from creation to disposal. Mankoff et al. explore how HCI practices can directly address sustainability, highlighting challenges in implementing energy-efficient design strategies[32]. Hanks et al.[19] present case studies on how small design changes in user interfaces can lead to

significant environmental benefits. Pierce et al.[36] focus on how everyday human practices can be rethought through HCI to promote more sustainable living. In line with those insights, Kim et al. [25] discuss design principles for ambient display that encourage sustainable behavior in users.

Similar studies can also be found in recent research. For instance, Issa and Isaias[45] propose sustainable design and argue that such insight must become central to the design of new technologies to make a concerted effort to tackle global environmental challenges. Several aspects are discussed, such as usability and user participation[30]. Preist [38] argues that to link the HCI is about the relationship between computers and society, then we must consider disciplines such as management science, economics, law, and policy studies, which consider the ways in which social institutions shape practices of both individuals and organisations. However, most HCI work does not take into account the complex relationships[26][43] between different stakeholders and technology[8][6], such as multi-disciplinary team in a specific project. This inevitably leads to the difficulty of achieving a design solution that can claim to promote sustainable development[27] or environmental protection simply through usability measurements, without balancing the demands of stakeholders[41]. Moreover, few HCI studies take on global complex environmental problems such as MPP, which is a missed opportunity for HCI to leverage its traditional knowledge of user interaction and contribute to environmental protection. A third gap observed by Dove et al.[12] is that HCI research rarely engage directly with municipal mitigation efforts, in other words decision-makers.

To foster an increased interest in technology design for sustainable development among stakeholders and to maximise the positive impacts of the tools, it is vital to explore the interplay between stakeholders, their actions towards sustainability, and their collaborative efforts in HCI research and its subfields. While numerous methods address sustainable design issues and aim to engage stakeholders in a cooperative effort [39], the essence lies in aligning more closely with stakeholders' needs and expectations. This alignment is achieved by harnessing their knowledge and experiences, using non-extractive strategies, including the use of boundary objects [16] and data as materials for design [11]. These principles are rooted in collaboration, co-creation, and empowerment.

Users are actively involved in the design process, offering feedback, proposing ideas, and taking part in decision-making. However, as our case study reveals, stakeholders involved in coastal clean-ups and management have diverse needs and tasks. Although the ultimate goal of most HCI studies is to create products and services that meet users' needs and help them achieve their goals [46] [37], as emphasized by Bannon in his seminal article "From human factors to human actors" [3], participants often have a unique understanding of their work environments and contexts. We should not confine our investigations to controlled settings, instead, we should explore real-world scenarios to understand the learning process [4], as we have done in this project, co-developing and using a visualization tool to support discussions on MPP. However, in this case, some stakeholders, the local and regional decision-makers, may not have the resources, knowledge or will to act on a global environmental challenge such as MPP. The tool designed therefore does not answer specific needs by the users, but acts as a nudging mechanism, providing the basis for decision-making.

We realised that one simple visualization software alone may never achieve the goal of promoting multi-party cooperation on the issue of MPP. However, without a platform to connect various stakeholders, we would not even have a dedicated arena for stakeholders to discuss the issue of MPP. Despite the challenges we faced in working with a multi-disciplinary team, our case provides valuable insights from various stakeholders and a comprehensive overview of the project's complexity [44] and the distinct needs of stakeholders. This collaborative approach allows all parties to benefit from each other's perspectives and contribute to our ongoing research.

The study is structured as follows: we first present the case in more detail in Section 2, highlighting the complexity of MPP and the challenges of designing a single design to support coastal clean-up operations and environmental decision-making. In Section 3, we discuss how the HCI field should reconsider the balance of power and interests when designing technology, detailing how we used a co-design approach to address the complexities of environmental protection tasks. Finally, in Section 4, we reflect on the key lessons learnt from our case study and suggest potential directions for future work on sustainability efforts in Norway.

2 The Case

2.1 Background

The project took place within the context of marine litter coastal clean-up initiatives in Møre and Romsdal, Western Norway. Marine Plastic Pollution (MPP) is a persistent challenge along the Norwegian coastline, which, due to its extensive length and exposure to ocean currents and winds, regularly accumulates litter from both coastal and maritime activities in the North Sea, Norwegian Sea and Barents Sea [31]. The litter is generally composed of various materials and originates from diverse sources (see Figure 1), which makes regulation and governance complex as the pollution transcends jurisdictional and economic boundaries.

Over the past decade, coastal communities in Norway have mobilized to form networks, bringing together stakeholders to tackle the issue of MPP through shared knowledge, resources, and coordinated clean-up efforts [21]. These ecosystems consist of organizations and individuals who share information, expertise, funding, and other resources to address the challenges of MPP (see Figure 2). One of their key missions is the organization of clean-up operations along the coast. These operations require information collection, assessment, and communication regarding the litter location, extent, and composition within the stakeholder ecosystem.

While some aspects of coastal clean-up can be facilitated by existing tools, much of the work still requires manual inspection and little is known of the overall plastic burden for Norway and the exact sources beyond theoretical models [15], which can limit action and policy-making at local, regional and national levels. To enhance data-driven support for these tasks, a project was initiated to create a visualization tool aimed at assisting various stakeholders in monitoring and decision-making on MPP. The project brought together a consortium of academic researchers, developers, clean-up volunteers and professional organizations as previously described



Figure 1: Photo of exhibition artifact illustrating the problem of marine plastic pollution and its impacts on wildlife, taken in Runde Research Centre, Møre and Romsdal, Norway



Figure 2: Various stakeholders involved in documenting and managing marine plastic pollution in Møre and Romsdal region, Norway

[21], but also recycling companies, and local and regional authorities in the Møre and Romsdal region (henceforth referred to as decision-makers).

The research outputs were, firstly, a visualization platform that dynamically and interactively mapped the clean-up efforts, the

quantities and the types of litter found around the region. Secondly, the development of models to recognize litter on photos, and thirdly, a model to identify potential sources by modelling litter movement at sea. The latter proved to be too imprecise, due to inadequate data; however, the first and second research outputs were achieved and explained in Wu et al.[48].

2.2 The workshop to designing the visualization tool

Throughout the spring of 2023, the research team took the initiative to develop a tool to support various stakeholders in collaborative efforts to fight Marine Plastic Pollution and local and regional levels as described previously (see Figure 3). The team engaged with these groups to understand their interests and needs, aiming to design a single tool, that would support all users involved in MPP management at local regional levels. The researchers took turns in presenting the developed prototypes and taking notes on the feedback and questions. The notes were then coded manually and analysed inductively.

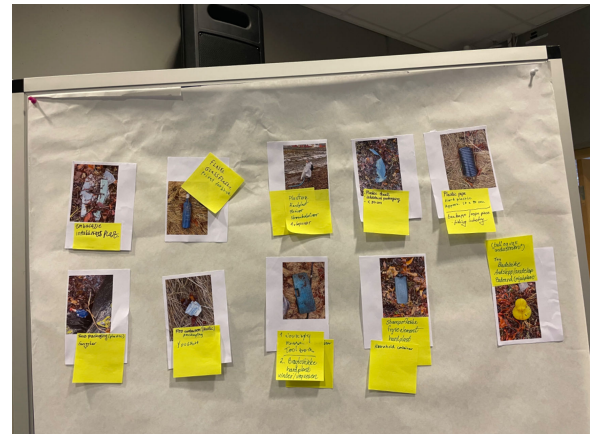


Figure 3: The outcomes of the joint activities of different stakeholders

The concept of the mobile application was that citizens could use the app to take photos of waste they encountered, and upload the images along with location data. The system would then automatically classify the waste and generate reports for other stakeholders to support their work. Later, drone technology was introduced to assist in monitoring areas that were inaccessible to people. Finally, a visualization tool would visualise the results both from an existing data repository that has litter collection data going back to 2013, and from the newly collected litter with the App as described in Wu et al. [48]. The stakeholders from the clean-up community agreed that such a system would be valuable, as it could provide a comprehensive overview of the coastline and help allocate resources more efficiently.

However, not all stakeholders shared the same needs or goals. For example, while parts of the system developed could help volunteer organisations identify the location and type of waste, this information was less relevant to decision-makers. Indeed, as we

learnt through the project, local and regional decision-makers are under no legal obligation to either monitor or take measures to address MPP, as it does not fall under any legislation and is not part of the Water Framework Directive. This also means that they do not have earmarked budget to tackle this issue and have to divide their limited budgets to best fulfill essential services they are obliged by the law to provide, such as nurseries, primary schools and health-related services. However, due to EU directives, they do have to tackle the issue of Circular Economy, which is to a certain extent linked to MPP as plastic fails to be reintegrated in the value chain and ends up in the environment [47].

Although the researchers sought to thoroughly understand the clean-up tasks even by engaging in the tasks themselves, the differing priorities among stakeholders complicated the project, making it challenging to design a tool that could support all stakeholders equally.

3 Case analysis

3.1 Technology tool for environmental protection is a boundary object

Our case is not unique; other studies have documented similar challenges with stakeholder needs in recent years [35] [49]. Although the visualization system was designed to meet the needs of all stakeholders, it has become evident that no single system can cater to the varied demands of every group. For example, for decision-makers, the visualizations lack the actionable insights necessary for their daily work, such as specific waste types and more importantly, the exact sources or at least industries and companies responsible for it. This is important for their work in promoting a circular economy and industrial symbiosis, where different private actors share resources and the waste from one can become the primary resource for another [14]. The visualizations fail to effectively communicate this information as the data does not have that level of detail, beyond "Fisheries" and "Industry". However, this is where Local Ecological Knowledge (LEK) plays a vital role as the volunteers and professionals who work on the ground to collect litter often know where the litter is likely to have come from, e.g., a ship, a factory nearby or households (Hellevik and Deshpande, 2023). The tool in itself does not transfer that knowledge, but workshops in which the tools are presented and tested, provide the space and arena for these stakeholders to meet and exchange this detailed knowledge.

On the other hand, waste collectors use a distinct classification system, where waste is categorized into material such as wood, metal, paint, hazardous materials, glass, and plastic, which determines which waste stream the items go to for recycling, incineration or landfill. This results in the same waste attributes being categorized differently, across stakeholder groups. One potential issue is the lack of standardized professional terminology; however, attempts are made at developing common terms at national and international levels (e.g., OSPAR Marine litter beach classification). The categories used by the volunteers in Norway are developed by the coordinating national organisations in Norway, namely 'Keep Norway Beautiful' (Hold Norge Rent) and the Marine Centre against Marine Litter (Senter mot Marin Forøpling, Marfo). These organisations define the normative context for the issue of MPP in Norway

and for our case. The categories used consists of 'fisheries', 'industry', 'personal use', 'hygiene' and 'other'. To streamline our tool with the current norms for the data collectors (i.e., the volunteers and professionals involved in clean-up operations), and to allow for synchronisation with existing data systems and avoid confusion or extra cognitive burden, we made the decision to align with this categorisation.

More importantly, this situation reflects Susan Star's view [16] that the visualization system itself is a boundary object [42]. The effectiveness of a visualization tool for different stakeholders depends on its use and interpretation. However, the flexibility of interpretation [16] can introduce a complex structure to the visualization. This complexity can make the tool more suited to a single stakeholder rather than a diverse group, and in such cases, the multi-dimensional functions of the tool may disappoint everyone if their specific needs are not met.

3.2 Data consequences of representational practices

The data itself presents another challenge. At the outset of the project, all stakeholders and researchers concurred that they could access public data and utilize citizens who collect waste items and take their own photographs. However, once the visualization platform was ready, the data raised significant concerns among all stakeholders. For instance, some stakeholders were concerned about the origins of the data and the synchronization of different data repository to avoid the duplication of work. They pointed out that errors in the data could render the technology, which is supposed to support their daily work, nonsensical. Moreover, they also questioned how the data was selected and based on what criteria and whether the people responsible for data recording were qualified and trained. Other stakeholders raised the same questions and went further to question the developers' confidence in the data accuracy. Although studies show that the accuracy of citizen science is generally high [1], there may be issues in the MPP domain if volunteers are untrained in spotting the litter as it may be buried, and in determining the categories. However, the high number of data points that citizen science opens up for means that individual inaccuracies usually even out.

All these concerns can be traced back to the discussion of boundary objects. This phenomenon also highlights the importance of the pragmatic programmer. In this domain, being a good listener is more crucial than being a good engineer. However, even with a good and responsive listener, the tools may still not be adopted by the users. MPP, similarly to other global environmental challenges, is a complex issue, even a "wicked problem" (Wagner, 2022) and decision-making processes on such topics are seldom linear, or rational, i.e., from evidence to action (Brunsson, 2007). On the contrary, decision-makers need to take into account many points of view and many aspects such as policy acceptance, economic growth, job creation and attractiveness for inhabitants, in addition to the evident environmental destructive impacts of MPP on our seas and coasts.

On a positive note, through this case, we have discovered that the tools we produce ultimately are less important than the process we go through to produce, test and evaluate them. Thus, it is essential to

connect with people and engage in the tasks that we aim to support to better understand the domain. In other words, the information representation[11]. It is easy to create an aesthetically pleasing visualization platform from the existing and newly-generated data, and we can readily add more features, believing we are giving others a decision-support. However, this case demonstrates that this does not necessarily mean the tool is useful[17]. We must integrate tools into the domain and integrate them with existing tools and systems, where the normative context demands it and with potential users.

The "data consequences prompt" is a contemplation of the limits of information visualization, and intriguingly, it examines what occurs at the edges of these data sets when they are represented in visualizations for various stakeholders. Our case study has taught us that confusion in utilizing data is not merely about integrating technology into a social context; it is about acknowledging the limits of marine litter categories and embracing flexible interpretation[11]. What remains unresolved is the misconception of the multifaceted nature of these data—understanding the underlying concepts and the additional features they bring to the visualization tool. This issue urgently requires solutions.

3.3 The need for multi-disciplinary teams

If we consider the stakeholders related to MPP and environmental protection in the form of coastal clean-up as a heterogeneous team, then researchers, especially system developers, must also be part of this team. However, within this project itself, system developers can also be divided into an inter-disciplinary team collaboration, that is, engineers and interaction designers and, as we are dealing with sustainability issues, natural and social scientists.

Indeed, the lack of domain knowledge in marine litter and oceanography led to misunderstandings and ideas based on unfounded assumptions. This gap was temporarily filled by an oceanographer who provided a lecture on ocean dynamics, in the first part of the PlastOPol project, and its effects on litter behaviour. However, knowledge of how volunteers carried out their clean-up tasks and recorded their results was also lacking. This gap was filled by 'becoming the user' as the entire research and design team joined several clean-up operations to develop experiential learning of marine litter clean up. This also made us realise the challenges in using a mobile App while cleaning up, due to being engaged with both hand pulling heavy or intricately buried litter from the soil. We further took part in several events organised by the volunteer network.

Unfortunately, besides engineers, other researchers, including designers themselves, often regard themselves as the end-product users, and hope that through the final testing, and iterating the system through test results, they can eventually produce a useful product. However, the obvious problem in multi-disciplinary team collaboration, especially in the field of technology product design, is that we believe interaction designers should play a leading role and take on the role of the planner for the entire system design[24].

However, our case show that a good interaction designer is not always trained to be a good development engineer. The compromise on leadership in the project ultimately results in what our case report shows: the needs of the multi-disciplinary team are not met, and the development results are not accepted by the team. But

how to make the multi-disciplinary teams work effectively? There is no good conclusion in the field of HCI, even though this issue is not new[24]. When people in HCI design and develop projects according to their strengths, such as interfaces, user experience, visualization results, etc., should system architecture, system integration, and requirement validation be included in HCI education and training? We do not know, but for our project, we feel this demand is badly needed.

4 Discussion

4.1 Design for all or design for one

Through this case, we deeply realized that it is extremely difficult to reach a consensus and design a visualization platform suitable for all stakeholders. This is not just a matter of how to meet the needs of different interest groups from a technical perspective, but more about how to balance their demands and reservations. Sometimes, designers can make some guiding design proposals to help stakeholders make trade-offs in their needs. However, in our case, each stakeholder has a very unique and professional perspective on the issue of marine waste classification, which makes the design itself complex, the system complex, and the interaction process complex, so that the final result cannot satisfy all stakeholders. We reflect that perhaps a multimodal interaction platform could solve such complex demand issues, but we more deeply realize that no single solution can solve all different problems. Designing a universal IT solution is challenging; however be more about planning, designing, and developing unique solutions that fit the current workflow of different groups is reachable. We noted that current HCI work is unilaterally pursuing strong generalizability, should we reflect on whether we have deviated in our approach? Simon[40] has emphasized paradigm shifts in understanding the natural and artificial worlds, and perhaps we do not need to create a new paradigm, can we truly reflect on how to turn to practice[29], taking needs and stakeholders' workflow in their daily life seriously and design useful systems?

4.2 Working in multi-disciplinary teams

Through this case, we have encountered a significant issue that is also a longstanding challenge in the field of HCI: the collaboration of multi-disciplinary teams. Randall [9] highlighted the cooperation issues between different teams in his book "Fieldwork for Design." However, his focus was primarily on the relationship between designers and a single final stakeholder group. In our scenario, designers had to work with developers and multiple stakeholders, often finding themselves in a less influential position. A design and social science perspectives were key in constantly reminding ourselves of the ultimate goals of providing useful and usable decision-support tools, and to evaluate the outputs against these goals, together with the project partners and volunteers. This was essential to avoid a shift of focus to technological feasibility rather than the ideal goal stated in the project. Knowledge transfer was crucial in both directions, however, as the goals of the project had to be adjusted to the technological feasibility and to the quality level of the data available.

We must clearly understand our role and contributions [28]. Without a clear sense of our identity, it becomes difficult to navigate conflicting requirements from different stakeholders. This not only complicates the process of balancing these demands but also hampers communication with developers. As we experienced, even with a well-intentioned approach such as co-design, the system design can fail due to minor challenges. For instance, our data sources, once a universally accepted solution, became a point of contention. This kind of team collaboration reveals that a one-size-fits-all solution is hard to achieve, even within the diverse stakeholders of our case. How do we break this impasse? Designing multiple plans for different stakeholders is one solution, but it inevitably extends the design cycle and increases the workload for designers and developers.

In the broader HCI community, there is ongoing discussion about HCI education[2]. Our case might offer some insights. Should HCI education[18] focus more on requirement analysis[13] or technical development? By doing so, could we avoid the wasted time spent discussing issues with different stakeholders from the outset? While this approach might sound more rational, could it be counterproductive in the design process?

4.3 Understanding the Normative Context

Another lesson learnt is that the normative context, consisting of laws and norms the users are expected to adhere to is crucial in determining the successful uptake of tools/motivation in using the tools. To understand this context, interviews can be carried out at the beginning of a project or individual meetings or focus groups with the potential users. Joining events is also useful in getting an insight into what norms they operate under. Indeed, as illustrated in the case of the litter categories, the 5 categories used by clean-up networks and organisations are the result of years of practice and is grounded in evidence of the most commonly found objects nationally. Changing these categories for our tools, would result in a misalignment and loss of relevance of our tools for the volunteers and decision-makers. On the other hand, attempting to change the national categories would involve lengthy processes and possibly result in a waste of resources on both sides.

4.4 Avoiding Extractive Practices and Fostering Long-Term Trust

The third lesson is that it is important to build good and long-term relationships with the stakeholders such as the volunteer organisations, and decision-makers, involving mutual respect and learning. Having the researchers/developers on an equal footing as them, or lower in the position of the learner to best develop tools that match the tasks and methods used or the type of information needed by the decision-makers in the format that is best for their needs and tasks. It is important to recognize the multitude of knowledge systems, including Local Ecological Knowledge (Hellevik and Deshpande, 2023), and knowledge about the legal and policy systems.

Indeed, as projects related to global challenges such as Marine Plastic Pollution are likely to span over long periods of time, it is important to maintain those positive long-term partnerships/relationships and to have one or more coordinators that can maintain or revive

the partnerships when needed. Good publicity and public awareness of the project and the tools can also contribute in expanding good relationships and give rise to new project collaboration when the apparatus is perhaps more receptive. As a continuation of this, it is good to be open to skepticism and criticism by the (potential) users, and those who have domain knowledge. It may also be important to not assume that the tool developed will be adopted or that the decision-makers will make use of it. As explained in point two, there may not be a legal frame within which to make decisions on that topic or in the way that the tool is suggesting.

4.5 The Power of Technology in Putting Global Challenges on the Policy Agenda

This leads to the last lesson learnt through the project, namely, that perhaps the most important impact that a tool can have is to put an important topic on the policy agenda at local and regional levels. Indeed, the buzz created around the hope of a technological fix, can stimulate more and more people to go from an attitude of defeatism to one of hope and motivation. Although natural scientists know that the challenges we face such as MPP will not be “fixed” through technology, but that deep societal and economic changes stimulated by strong regulations will be unavoidable, many still place hope in information systems and other forms of technology to provide quick and pain-free solutions. It can be debated whether this should be encouraged or used at all and in our case, we only realized this after the project was well underway. It is crucial for HCI and other technology developers to always be honest about what the technology can and, especially, what it cannot deliver. However, putting MPP on the agenda of local and regional policy-makers, where it previously was not and helping to secure more funding to continue the difficult work of cleaning our coast is a proud achievement of our project team.

5 Conclusion

This paper reports on a unique case of multi-stakeholder participation in marine litter, analyzes the case itself, including both successful and unsuccessful attempts, and reflects on how to handle the design of visualization systems for unique fields with multiple stakeholders, as well as some discussions that the HCI field may benefit from. Although this topic is ongoing, we hope that through our lessons learnt, we can inspire HCI researchers who are committed to in-the-wild studies to tackle global environmental challenges and sustainability challenges through practical application of HCI topics.

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